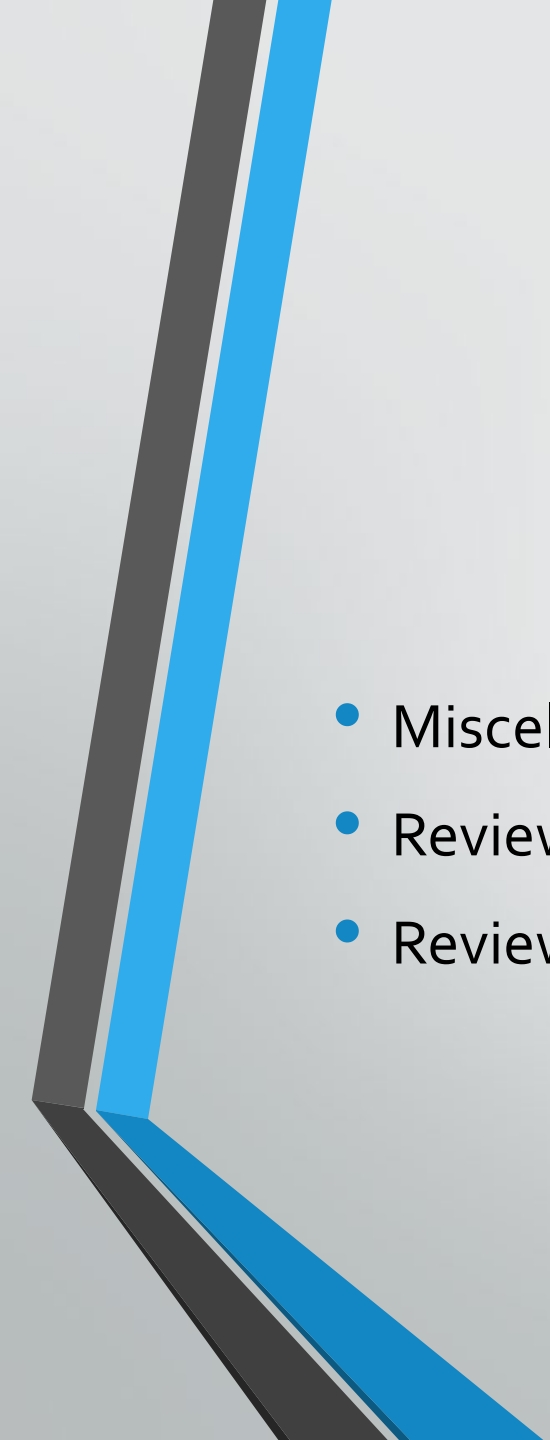




Statistical Methods & Machine Learning 1

Class #6

John Kubale
October 1, 2024



Outline

- Miscellaneous points
- Review from previous lecture and HW₄
- Review

Exam 1

- All materials will be available via the Class 6 module starting at 12pm today and will be due by the start of class next Tuesday (930am).
- The exam will follow a similar layout to your homeworks.
 - Complete and submit a knitted pdf that contains your name at the top, the numbered questions, and your answers (this includes all relevant code and output).
 - **Please check and remove any unnecessary output before submission.**
- Must complete the exam on your own.
 - Discussion of the exam with others and use of AI/LLMs is not allowed.
 - Use of notes, class materials, and the internet (within reason) is permitted
 - i.e., you may not post exam questions on forums to solicit help

Exam 1

- The first question largely requires you to correctly interpret output as you will not have access to the data.
 - While you won't have access to the data, you will have all the information needed to answer the question.
 - R can be helpful with some of the calculations required.
 - You will need to include your answers in R so they're in the knitted pdf anyway.
- The second question will involve actually working with a dataset.

Tips on knitting to a pdf

```
1 ▾ ---  
2 title: "Homework 4 Solution"  
3 author: "John Kubale"  
4 date: "2024-09-23"  
5 output: pdf_document  
6 number_sections: yes  
7 fontsize: 12pt  
8 fig_width: 7  
9 fig_height: 6  
10 ▲ ---
```

- Put your name at the top using the author field in the header
- Avoid copy/pasting from other documents!

Review – Lecture 5

- Ambiguous quiz question from last week:

1

Multiple Choice 1 point

For a given linear regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, what is true about ϵ_i ?

- ☐ ϵ_i increases as the outcome variable increases.
- ☒ ϵ_i contains information about how well the model represents the data.
- ☐ ϵ_i is known because of OLS.
- ☐ OLS attempts to minimize ϵ_i .

Credit given for
either of these
answers

Review – Lecture 5

- Ambiguous quiz question from last week:

For a given linear regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, what is true about ϵ_i ?

- ☐ ϵ_i increases as the outcome variable increases.
- ☒ ϵ_i contains information about how well the model represents the data.

Correct
Answer:

OLS attempts to minimize ϵ_i .

- ☐ ϵ_i is known because of OLS.
- ☐ OLS attempts to minimize ϵ_i .

Not the correct answer

Review – Lecture 5

- F tests and the null hypotheses being tested:
 - `summary(model)`
 - `anova(model)`
 - `anova(model1, model2)`

Review – Lecture 5

- Let's fit a simple linear regression model using the `mtcars` dataset in R. Specifically we'll model `mpg` as a function of `hp`.

```
data("mtcars")

m1 <- lm(mpg ~ hp, mtcars)
summary(m1)

##
## Call:
## lm(formula = mpg ~ hp, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7121 -2.1122 -0.8854  1.5819  8.2360
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  30.09886    1.63392   18.421  < 2e-16 ***
## hp          -0.06823    0.01012   -6.742  1.79e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.863 on 30 degrees of freedom
## Multiple R-squared:  0.6024, Adjusted R-squared:  0.5892
## F-statistic: 45.46 on 1 and 30 DF, p-value: 1.788e-07
```

What is the null hypothesis for the F statistic here?

Review – Lecture 5

- What about when we use the `anova()` function on the same model object?

```
anova(m1)
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## hp           1  678.37   678.37   45.46 1.788e-07 ***
## Residuals  30  447.67    14.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

What is the null hypothesis for the F statistic here?

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```
data("mtcars")
```

```
m1 <- lm(mpg ~ hp, mtcars)
summary(m1)
```

```
##
## Call:
## lm(formula = mpg ~ hp, data = mtcars)
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Review – Lecture 5

- What about when we use the `anova()` function on the same model object?

```
anova(m1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: mpg
```

```
##      Df Sum Sq Mean Sq F value
```

```
## hp      1 678.37   678.37  45.46 1.7
```

```
## Residuals 30 447.67   14.92
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

They're the same test. You're comparing your fit model (m1), which here has 1 predictor to that of the intercept only model (0 predictors). When you don't specify the model to compare to R assumes that you want to compare to the intercept only model.

```
## Coefficients:
```

```
##      Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 30.09886    1.63392  18.421 < 2e-16 ***
```

```
## hp      -0.06823    0.01012  -6.742 1.79e-07 ***
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```
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```

```
##
```

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```
## F-statistic: 45.46 on 1 and 30 DF, p-value: 1.788e-07
```

Review – Lecture 5

How would we write this out?

- $H_0: \beta_1 = \beta_2 = \beta_3 = 0$
- $H_0: RSS_{Full} = RSS_{Reduced}$
- Here our reduced model is the intercept only model.
- What would you conclude based on this test?

Review – Lecture 5

Let's now add a second predictor to our model and compare it to our simple linear regression model using an F test.

```
m2 <- lm(mpg ~ hp + wt, mtcars)
anova(m1, m2)

## Analysis of Variance Table
##
## Model 1: mpg ~ hp
## Model 2: mpg ~ hp + wt
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      30 447.67
## 2      29 195.05  1    252.63 37.561 1.12e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

What is the null hypothesis for the F statistic here?

Review – Lecture 5

Let's now add a second predictor to our model and compare it to our simple linear regression model using an F test.

The null hypothesis here is nearly the same as the one we saw in the previous example. It still states that the RSS of the full model is equal to that of the reduced model—the difference is what reduced model is being used. Before it was an intercept only model, while here it is a simple linear regression model.

```
m2 <- lm(mpg ~ wt, data = mtcars)
anova(m1, m2)

## Analysis of Variance
##
## Model 1: mpg ~ 1
## Model 2: mpg ~ wt
## Res.Df    RSS    Df Sum of Sq    F    Pr(>F)
## 1      30 447.67    1      252.63 37.561 1.12e-06 ***
## 2      29 195.05    1      252.63 37.561 1.12e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Review – Lecture 5

How would we write this out?

- $H_0: RSS_{Full} = RSS_{Reduced}$
- Here our reduced model is the simple linear regression model.
- What would you conclude based on this test?

HW₄

- Let's look at questions from the homework you just turned in.



Exam Review

Review – Regression terminology

- The model $\text{lm}(y \sim x + z, \text{data})$ can be described as:
 - Regress y on x and z
 - Fit a linear model of y as a function of x and z

Review – Degrees of freedom

- Can be calculated using the formula $n - p$
 - $n = \text{sample size}$
 - $p = \text{number of parameters being estimated}$
- With regression models sometimes will sometimes see it written as $n - p - 1$ with 1 representing the intercept → since the intercept is a parameter too this is accounted for by p

Review – Degrees of freedom

- You have a dataset ($n=50$) with 3 variables (y , x , and z).
- How many degrees of freedom do you have in each of the following models?
 - $\text{lm}(y \sim 1, \text{data})$
 - $\text{lm}(y \sim x, \text{data})$
 - $\text{lm}(y \sim z, \text{data})$
 - $\text{lm}(y \sim x + z, \text{data})$



Interpretations

Review – Intercept Only Regression Interpretations

- Given the intercept only linear regression model, $Y = \beta_0 + \epsilon$
 - The coefficient for β_0 represents the expected mean value of the outcome, Y .
 - As there are no predictors this should be equivalent to the arithmetic mean.

Review – Simple Linear Regression Interpretations

- Given the simple linear regression model, $Y = \beta_0 + \beta_1 x_1 + \epsilon$
 - The coefficient for β_0 represents the expected mean value of the outcome, Y , when $x = 0$ (or its reference group).
 - The coefficient for β_1 represents the change in the expected mean value of the outcome, Y , for a one-unit increase in X .

Review – Multiple Linear Regression Interpretations

- Given the multiple linear regression model, $Y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \epsilon$
 - The coefficient for β_0 represents the expected mean value of the outcome, Y , when both x_1 and x_2 equal 0 (or their reference group).
 - The coefficient for β_1 represents the change in the expected mean value of the outcome, Y , for a one-unit increase in x_1 while holding x_2 constant.
 - The coefficient for β_2 represents the change in the expected mean value of the outcome, Y , for a one-unit increase in x_2 while holding x_1 constant.

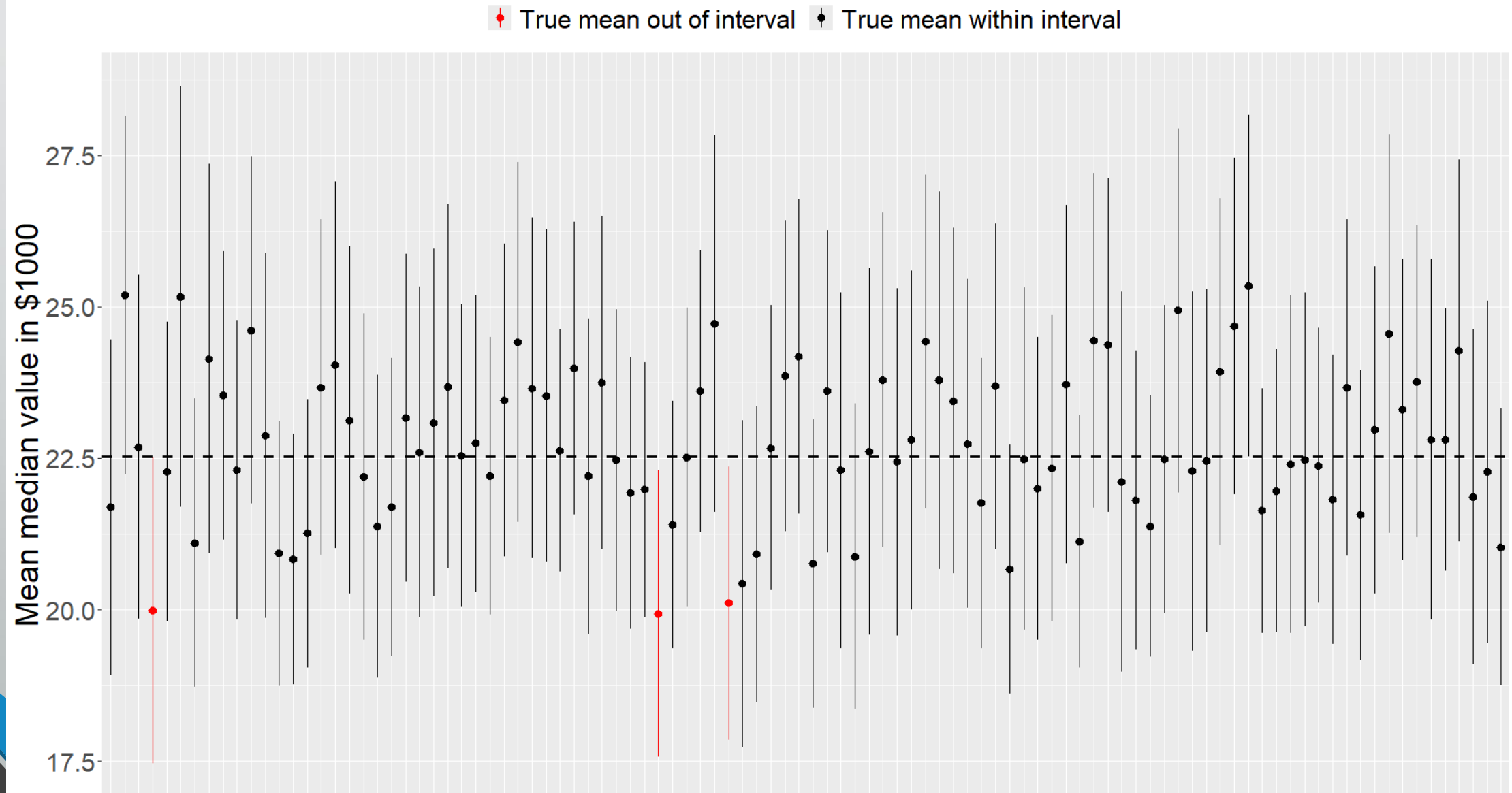
Review – 95% CI Interpretation

- Given the following 95% CI for a regression coefficient (i.e., beta), how would you interpret it?
 - 95% CI (1.2, 2.4)

Review – 95% CI Interpretation

- Given the following 95% CI for a regression coefficient (i.e., β), how would you interpret it?
 - 95% CI (1.2, 2.4)
- 1. If we were to take repeated samples from the population, fit the model on each, and estimate 95% CI we would expect the true population β to fall within the interval 95% of the time.

Review – 95% CI Interpretation



Review – 95% CI Interpretation

- Given the following 95% CI for a regression coefficient (i.e., β), how would you interpret it?
 - 95% CI (1.2, 2.4)
- 1. If we were to take repeated samples from the population, fit the model on each, and estimate 95% CI we would expect the true population β to fall within the interval 95% of the time.
- 2. We are 95% confident that the true population β is between 1.2 and 2.4.



Calculating “by hand”

Review – Calculating “by hand”

- You may be asked to calculate things “by hand” on this or other exams.

Confidence intervals

- You might have the data to work with in R in which case you would need to manually obtain the appropriate test statistic (e.g., using the `qt()` function), but could use R functions to obtain other key components like the SE.
- You might not have access to the data (e.g., you only have the output).
 - Then you would need to identify the necessary output from the information given and use it to get the needed quantities to estimate the 95% confidence interval

Review – Calculating “by hand”

Confidence intervals

- What information do you need to calculate a confidence interval?
 - Quantity the interval corresponds to (mean, beta, etc.).
 - Standard error of the quantity
 - Test statistic

Review – Calculating “by hand”

Confidence intervals

- What information do you need to calculate a confidence interval?
 - Quantity the interval corresponds to (mean, beta, etc.).
 - Standard error of the quantity
 - Test statistic (be careful you're using the correct one)
- Where might you look for these in output?
- How could you obtain them if you have access to the data?

Review – Calculating “by hand”

- Could be asked to calculate other quantities “by hand”.
 - All will be things you’ve already seen! → look at your labs and homeworks

Review – Comparing models

- You may be asked to compare models and state (with justification) which you would prefer.
 - This could be when you have access to the data, or only some output.
- What might you use?

Review – Comparing models

- You may be asked to compare models and state (with justification) which you would prefer.
 - This could be when you have access to the data, or only some output.
- What might you use?
- What are some potential limitations of these methods that we've discussed?